

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
SAN DIEGO REGION**

TENTATIVE ADDENDUM NO. 3 TO ORDER NO. 97-52

**AN ADDENDUM ADDING THE GOREN WASTEWATER TREATMENT PLANT TO THE
WASTE DISCHARGE AND RECYCLING REQUIREMENTS FOR THE
SOUTH ORANGE COUNTY WASTEWATER AUTHORITY
ORANGE COUNTY**

The California Regional Water Quality Control Board, San Diego Region (hereafter Regional Board) finds that:

1. Order No. 97-52, *Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Wastewater Authority* prescribes requirements for the South Orange County Wastewater Authority (SOCWA) and its member agencies listed in Table I of the Order.
2. Trabuco Canyon Water District (TCWD) is a SOCWA member agency and will operate the Goren Wastewater Treatment Plant (GWTP) that will be built on Parcel 856-042-24 near Summit Trail Road and Live Oak Canyon Road, Trabuco Canyon in the Upper Trabuco Hydrologic Subarea (HAS 901.22).
3. The requirements of Order No. 97-52 are consistent with the Water Quality Control Plan for the San Diego Basin (9) (hereafter Basin Plan).
4. The GWTP will treat and discharge up to 7,000 gallons per day or approximately 7 acre-feet of domestic wastewater per year on open space within the Oaks at Trabuco, which is a proposed nine-lot residential development located within the Upper Trabuco Hydrologic Subarea. The proposed discharge is consistent with the provisions of Order No. 97-52 and the Basin Plan.
5. The proposed GWTP will consist of flow equalization, sequencing batch reactor, coagulation, filtration, and disinfection with sodium hypochlorite. The GWTP will be designed, constructed and operated in a manner that complies with all applicable Title 22 water recycling regulations.
6. The GWTP will include an on-site 20,000 gallon storage tank, and if necessary during wet weather periods, TCWD will haul recycled water from the storage tank to the District's sanitary sewer system or water reclamation plant.
7. The proposed discharge will not cause the SOCWA member agencies to exceed the annual limit specified in Order No. 97-52 for volume of recycled water that may be discharge within the Upper Trabuco Hydrologic Subarea.

8. The Regional Board has notified all known interested parties of its intent to amend waste discharge and recycling requirements.
9. On January 17, 1996 the Orange County Subdivision Committee found that a Final EIR, certified by the County of Orange on December 10, 1991, and Addendum No. IS 91106 are adequate to satisfy the requirements of CEQA for this project.
10. The Regional Board has considered all water resource related environmental factors associated with the proposed discharge of waste.
11. The Regional Board in a public meeting heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED, that:

1. Table 5 of Order No. 97-52 is amended to include the Goren Wastewater Treatment Plant as a facility under TCWD with a proposed discharge of 0.007 million gallons per day.
2. The following requirement is added to Order No. 97-52 as Facility Design and Operation Specification C.9:

C.9 CERTIFICATION REPORT

The Goren Wastewater Treatment Plant and disposal facilities shall be completely constructed and operable prior to the initiation of the discharge and the complete facilities shall have adequate capacity for the full design flow of 0.007 million gallons per day (MGD) average daily flow and 0.0175 MGD peak instantaneous flow. A report from the design engineer certifying the adequacy of each component of the treatment, storage, and disposal facilities shall be submitted by the discharger prior to commencement of the discharge. The certification report shall contain a requirement-by-requirement analysis based on acceptable engineering practices, of how the process and physical designs of the facilities will ensure compliance with the master reclamation permit. The certification report shall also include an inspection schedule for the GWTP to ensure runoff does not occur. The design engineer shall affix their signature and engineering license number to the certification report and should submit it prior to construction of the facilities. Recycled water shall not be purveyed to a user until all of the following have occurred:

- a. The certification report is received and accepted as complete by the Regional Board.
- b. The Regional Board has been notified that the Title 22 report and the rules and regulations for recycled water reuse are approved by the State DHS and County DEH,

- c. The Regional Board has been notified of the completion of facilities by Trabuco Canyon Water District,
 - d. An inspection of the facilities has been made by the Regional Board, and
 - e. The Regional Board notifies the Recycled Water Agency by letter that the discharge can be initiated.
3. The following requirement is added to Order No. 97-52 as Facility Design and Operation Specification C.10:

C.10 ENGINEERING REPORT

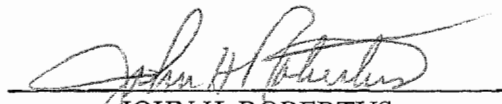
Prior to discharge, the Goren Wastewater Treatment Plant shall meet the design, operational, and reliability requirements of Articles 7, 8, 9 and 10 of the California Code of Regulations, Title 22, Division 4, Chapter 3. Trabuco Canyon Water District shall prepare an engineering report conforming to the California Code of Regulations, Title 22, Engineering Report Guidelines, Sections 3 and 4. The engineering report shall be submitted to the State DHS, County DEH, and the Regional Board.

4. The following requirement is added to Monitoring and Reporting Program No. 97-52 as Provision J:

Provision J. Goren Wastewater Treatment Plant – Disposal Log

The Trabuco Canyon Water District shall submit an annual report listing the dates, volumes and disposal locations for biosolids, recycled water, and wastewater discharged or transported from the Goren Wastewater Treatment Plant.

I, John H. Robertus, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Addendum adopted by the California Regional Water Quality Control Board, San Diego Region, on September 8, 2004


JOHN H. ROBERTUS
Executive Officer

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
SAN DIEGO REGION**

**ADDENDUM NO. 2
TO
ORDER NO. 97-52**

**WASTE DISCHARGE AND WATER RECYCLING REQUIREMENTS
FOR THE
SOUTH ORANGE COUNTY RECLAMATION AUTHORITY
ORANGE COUNTY**

The California Regional Water Quality Control Board, San Diego Region (hereinafter Regional Board), finds that:

1. On October 15, 1997, this Regional Board adopted Order No. 97-52, Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of South Orange County Reclamation Authority. Order No. 97-52 prescribes requirements for the South Orange County Reclamation Authority and its member agencies listed in Table 1 of the Order.
2. Finding No. 1 of Order No. 97-52 identifies Los Alisos Water District as a member agency and Irvine Ranch Water District as an associate member agency, of the South Orange County Reclamation Authority.
3. Los Alisos Water District and Irvine Ranch Water District have merged. Irvine Ranch Water District has assumed all responsibilities of both agencies with respect to sewage collection, treatment, disposal, and compliance with the requirements of Order No. 97-52.
4. On May 24, 2001, the Regional Board was notified that the South Orange County Reclamation Authority, Aliso Water Management Agency, and the South East Regional Reclamation Authority will consolidate their operations, effective July 1, 2001, into a single new Joint Powers Authority, the South Orange County Wastewater Authority.
5. Consolidation of the aforementioned entities that are subject to regulation under Order No. 97-52 is exempt from the provisions of the California Environmental Quality Act (Public Resources Code, Section 21000, et seq.) because it involves a change in the organization of local governmental entities, in accordance with Section 15320, Chapter 3, Title 14, California Administrative Code.
6. The Regional Board has notified the dischargers identified in Order No. 97-52 and all known interested parties of its intent to amend Order No. 97-52.
7. The Regional Board has, at a public meeting on October 10, 2001 held or provided an opportunity for a public hearing, and heard and considered all comments pertaining to the terms and conditions of this addendum.

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IT IS HEREBY ORDERED, that:

1. Finding No. 1 of Order No. 97-52 is amended to delete Irvine Ranch Water District as an associate member agency from the list of South Orange County Reclamation Authority member agencies.
2. All references to the Los Alisos Water District in Order No. 97-52 and addenda thereto shall henceforth refer to Irvine Ranch Water District.
3. All references to the South Orange County Reclamation Authority in Order No. 97-52 and addenda thereto shall henceforth refer to the South Orange County Wastewater Authority as the entity subject to regulation under Order No. 97-52.

I, John H. Robertus, Executive Officer, do hereby certify the foregoing is a full, true and correct copy of an Addendum adopted by the California Regional Water Quality Control Board, San Diego Region, on October 10, 2001.



JOHN H. ROBERTUS
Executive Officer

California Regional Water Quality Control Board
San Diego Region

ADDENDUM NO. 1 TO ORDER NO. 97-52

AN ADDENDUM ADDING THE SOUTH COAST WATER DISTRICT TO THE
WASTE DISCHARGE AND WATER RECYCLING REQUIREMENTS
FOR THE
SOUTH ORANGE COUNTY RECLAMATION AUTHORITY,
ORANGE COUNTY

The California Regional Water Quality Control Board, San Diego Region (hereinafter Regional Board) finds that:

1. Order No. 97-52, *Waste Discharge and Water Recycling Requirements for the Production and Purveyance of Recycled Water by Member Agencies of the South Orange County Reclamation Authority* prescribes requirements for the South Orange County Reclamation Authority (SOCRA) and its member agencies listed in Table 1 of the Order.
2. Order No. 97-19 *Waste Discharge and Water Recycling Requirements for the South Coast Water District Advanced Wastewater Treatment Plant, Orange County* prescribes requirements for the treatment and disposal of recycled water by the South Coast Water District.
3. This Addendum adds South Coast Water District, who became a member agency of SOCRA in August 1999, to Order No. 97-52 and rescinds Order No. 97-19.
4. The Regional Board has notified all known interested parties of its intent to amend waste discharge and water recycling requirements.
5. On September 10, 1992, South Orange County Reclamation Authority approved a Negative Declaration for the use of recycled water within the SOCRA jurisdictional area in accordance with the California Environmental Quality Act (Public Resources Code Section 21000, et seq.), after a public hearing was held. The project as approved by SOCRA will not have a significant effect on water quality.
6. The Regional Board has considered all water resource related environmental factors associated with the proposed discharge of waste.
7. The Regional Board in a public meeting heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED, that:

1. Table 1 and Table 2 of Order No. 97-52 are superceded by the following:

Table 1

SOCRA MEMBER AGENCIES

AGENCIES	MEMBERSHIP STATUS
Capistrano Valley Water District	MEMBER
El Toro Water District	MEMBER
Irvine Ranch Water District	ASSOCIATE MEMBER
Los Alisos Water District	MEMBER
Moulton Niguel Water District	MEMBER
Santa Margarita Water District	MEMBER
South Coast Water District	MEMBER
Trabuco Canyon Water District	MEMBER

Table 2

SANITATION AGENCIES WITHIN SOCRA

AGENCIES
El Toro Water District
Emerald Bay Service District
City of Laguna Beach
Los Alisos Water District
Moulton Niguel Water District
City of San Clemente
City of San Juan Capistrano
Santa Margarita Water District
South Coast Water District
Trabuco Canyon Water District

2. Discharge Specification A.1 is superceded by the following:

A.1. The discharge for landscape irrigation and other land disposal projects of an effluent containing pollutants in excess of the limitations specified in Table A-1 and Table A-2 is prohibited.

Table A-1
EFFLUENT LIMITATIONS FOR RECYCLED WATER
COMMON TO ALL FACILITIES WITHIN THE SOCRA AREA

CONSTITUENT	12 month ¹ Avg. (mg/L)	Daily ² Maximum (mg/L)
Biochemical Oxygen Demand (BOD ₅ @ 20°C)	30	45
Total Suspended Solids	30	45
pH (within limits shown at all times)	6.0 - 9.0	6.0 - 9.0
Total Dissolved Solids	1000 ³	1100 ³
Chloride	375 ³	400 ³
Sulfate	400 ³	450 ³
% Sodium	60	65
Nitrate	45	50
Iron	0.3	0.4
Manganese	0.05	0.06
MBAS	0.5	0.6
Boron	0.67	0.75
Odor	NONE	NONE
Aluminum	1.0	
Arsenic	0.05	
Antimony	0.006	
Barium	1.0	
Beryllium	0.004	
Cadmium	0.005	
Chromium	0.05	
Cyanide	0.2	
Mercury	0.002	
Nickel	0.1	
Selenium	0.05	
Thallium	0.002	

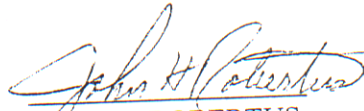
- 1 The 12 month average effluent limitation shall apply to the arithmetic mean of the results of all samples collected during any 12 month consecutive calendar period.
- 2 The daily maximum effluent limitation shall apply to the results of a single composite or grab sample.
- 3 Concentrations of total dissolved solids, chloride and sulfate in the discharge of effluent from SCWD for landscape irrigation and other land disposal projects within H.S.A. 901.13, 901.14 and 901.27 shall not exceed the limitations specified in Table A-2.

Table A-2
TDS, CHLORIDE AND SULFATE LIMITATIONS
SOUTH COAST WATER DISTRICT

CONSTITUENT	12 month ¹ Avg. (mg/L)	Daily ² Maximum (mg/L)
Total Dissolved Solids	1200	1250
Chloride	400	450
Sulfate	500	550

3. Order No. 97-19 *Waste Discharge and Water Recycling Requirements for the South Coast Water District Advanced Wastewater Treatment Plant, Orange County* is rescinded.

I, John Robertus, Executive Officer, do hereby certify the foregoing is a full, true and correct copy of an Addendum adopted by the California Regional Water Quality Control Board, San Diego Region, on October 13, 1999.


JOHN H. ROBERTUS
Executive Officer

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
SAN DIEGO REGION**

**ORDER NO. 97-52
WASTE DISCHARGE AND WATER RECYCLING REQUIREMENTS
FOR THE PRODUCTION AND PURVEYANCE OF RECYCLED WATER BY
MEMBER AGENCIES OF THE
SOUTH ORANGE COUNTY RECLAMATION AUTHORITY,
ORANGE COUNTY**

The California Regional Water Quality Control Board, San Diego Region (hereinafter Regional Board), finds that:

1. On September 12, 1992, this Regional Board adopted Order No. 92-67, *South Orange County Reclamation Authority, Orange County, Consolidation of Requirements*. Order No. 92-67 prescribed recycled water production and purveyance requirements to the South Orange County Reclamation Authority (SOCRA) and its member agencies. The current membership of SOCRA is shown in Table 1.

Table 1

SOCRA MEMBER AGENCIES

AGENCIES	MEMBERSHIP STATUS
Capistrano Beach Water District	MEMBER
Capistrano Valley Water District	MEMBER
El Toro Water District	MEMBER
Irvine Ranch Water District	ASSOCIATE MEMBER
Los Alisos Water District	MEMBER
Moulton Niguel Water District	MEMBER
Santa Margarita Water District	MEMBER
South Coast Water District	ASSOCIATE MEMBER
Trabuco Canyon Water District	MEMBER

The institutional boundaries of SOCRA and its member agencies are shown in Figure 1.

2. On April 28, 1997, David A. Caretto, General Manager of SOCRA, submitted a Report of Waste Discharge in application for modified waste discharge requirements. The requested modifications of Order No. 92-67 included increased production and reuse volumes, changes to the Monitoring and Reporting Program and changes to effluent limitations. The application was determined complete on July 1, 1997.

3. This Order regulates discharges of recycled water within the service areas of all member agencies of SOCRA, except for the South Coast Water District (SCWD). Discharges within the SCWD service area are regulated under Order No. 97-19, *Waste Discharge and Water Recycling Requirements for the South Coast Water District Advanced Wastewater Treatment Plant, Orange County*. South Coast Water District has an agreement with Moulton Niguel Water District to provide up to 1000 acre-feet per year to Moulton Niguel Water District for distribution.
4. El Toro Water District (ETWD) produces secondarily treated recycled water. Recycled water from ETWD is delivered to one user, Professional Community Management. The recycled water is used for irrigation purposes. Professional Community Management lies entirely within the Santa Ana Regional Board jurisdiction. El Toro Water District is regulated under Order No. 94-03, issued by the Santa Ana Regional Board. Secondarily treated recycled water is not provided to SOCRA for distribution within this Regional Board's jurisdiction.
5. The sanitation agencies shown in Table 2 are responsible for the collection, treatment, and/or disposal of sewage within SOCRA.

Table 2

SANITATION AGENCIES WITHIN SOCRA

AGENCIES
Capistrano Beach Water District
Dana Point Sanitary District
El Toro Water District
Emerald Bay Service District
City of Laguna Beach
Los Alisos Water District
Moulton Niguel Water District
City of San Clemente
City of San Juan Capistrano
Santa Margarita Water District
South Coast Water District
Trabuco Canyon Water District

6. The Aliso Water Management Agency (AWMA) is a joint powers authority formed to operate regional sewage treatment and disposal facilities, including an

ocean outfall, on behalf of its member agencies within the Aliso watershed. Table 3 lists the AWMA member agencies.

Table 3

ALISO WATER MANAGEMENT AGENCY MEMBERS

AGENCIES
El Toro Water District Emerald Bay Service District City of Laguna Beach Los Alisos Water District Moulton Niguel Water District South Coast Water District

7. The South East Regional Reclamation Authority (SERRA) is a joint powers authority formed to operate regional sewage treatment and disposal facilities, including an ocean outfall, for the benefit of its member agencies, as shown in Table 4. SERRA encompasses that area of south Orange County north of the Orange/San Diego County boundary and south of the AWMA jurisdiction.

Table 4

SOUTH EAST REGIONAL RECLAMATION AUTHORITY MEMBERS

AGENCIES
Capistrano Beach Water District Dana Point Sanitary District Moulton Niguel Water District City of San Clemente City of San Juan Capistrano Santa Margarita Water District

8. The City of San Clemente is not a member of SOCRA. However, the City of San Clemente may provide recycled water to SOCRA for distribution within the SOCRA service area. The City of San Clemente's Water Reclamation Plant is regulated under Order No. 91-50, *Waste Discharge Requirements for the City of San Clemente Water Reclamation Plant, Reclamation Projects, Orange County*.
9. Table 5 lists the existing Title 22 capacity and production and planned future Title 22 capacity for each SOCRA member agency and for SERRA and the City of San Clemente, which may provide recycled water to the SOCRA service area in the future.
10. Table 6 indicates the current annual volume and projected ultimate annual volume of recycled water for use in each hydrologic subunit within SOCRA.

Table 5

EXISTING AND PROPOSED RECYCLED WATER PRODUCTION
FOR SOCRA SERVICE AREA

AGENCY	FACILITY	SOCRA SERVICE AREA TITLE 22 CAPACITY ^{A,B}			
		EXISTING ORDER	CURRENT PERMITTED WASTE DISCHARGE (MGD)	CURRENT PRODUCTION (MGD) ^J	PROPOSED WASTE DISCHARGE ^H (MGD)
CBWD	CBWD WPCF	92-67	2.80	0.00	2.80
SERRA	SERRA WPCF	NONE	0.00	0.00	7.00
ETWD	El Toro AWT	94-03 ^C	2.60 ^{D,E}	0.33 ^{D,E}	2.60 ^{D,E}
IRWD	Michelson WRP	92-67	0.00	0.00	0.00
LAWD	Los Alisos AWT	94-03 ^C	5.50 ^{E,F}	1.27 ^E 0.13 ^F	5.50 ^{E,F}
MNWD	Joint Regional Plant Plant 3A WRP AWMA Coastal WRP	92-67	4.80	0.84	11.40
		92-67	2.40	0.79	5.20
		92-67	0.00	0.89 ^G	0.89 ^G
CSC	San Clemente WRP	91-50	0.00	0.79 ^I	4.40 ^I
SMWD	Chiquita WRP Oso Creek WRP Nichols Institute	92-67	6.00	0.00	7.50
		92-67	3.00	0.00	3.00
		92-67	0.08	0.02	0.08
SCWD	SCWD WRP	97-19	0.00 ^G	0.89 ^G	0.89 ^G
TCWD	Robinson Ranch WRP	92-67	1.10	0.65	1.10

Footnotes:

- A Tertiary disinfected effluent unless otherwise noted; produced in and for Region 9 unless otherwise noted.
- B Capacities expressed as 30-day average.
- C Order adopted by RWQCB, Santa Ana Region (Region 8).
- D Secondary disinfected effluent.
- E Santa Ana Region (Region 8).
- F San Diego Region (Region 9).
- G Agreement between SCWD and MNWD for SCWD to provide up to 1000 AF/yr to MNWD.
- H These values are planned or proposed increases in flow during the 5 year permit period.
- I The City of San Clemente is not currently providing recycled water to the SOCRA service area, however, the City may provide recycled water to SOCRA and may receive recycled water from SMWD in the future.
- J 1996 values.

Table 6

EXISTING AND PROPOSED RECYCLED WATER USE
BY HYDROLOGIC SUBUNIT FOR SOCRA SERVICE AREA

HYDROLOGIC SUBUNIT	HYDROLOGIC NAME	CURRENT PERMITTED VOLUME OF USE (AC-FT/YR)	CURRENT VOLUME OF USE (AC-FT/YR) ^A	PROPOSED VOLUME OF USE (AC-FT/YR)
REGION 8		6129	2428	6129
1.10	LAGUNA HA			
1.11	San Joaquin Hills HSA	0.00	0.00	0.00
1.12	Laguna Beach HSA	502	0.00	1026
1.13	Aliso HSA	7330	1280	10494
1.14	Dana Point HSA	4280	1359	5804
1.20	MISSION VIEJO HA	19349	5081	
1.21	Oso HSA			7168
1.22	Upper Trabuco HSA			420
1.23	Middle Trabuco HSA			4232
1.24	Gobernadora HSA			4148
1.25	Upper San Juan HSA			977
1.26	Middle San Juan HSA			0.00
1.27	Lower San Juan HSA			4396
1.28	Ortega HSA			2758
1.30	SAN CLEMENTE HA	3890	706	3890
1.40	SAN MATEO CANYON HA	900	265	837
SOCRA		36251	11119	52279

A 1996 Values.

11. Definitions:

AWMA - Aliso Water Management Agency

CCR - California Code of Regulations

CBWD - Capistrano Beach Water District

CEQA - California Environmental Quality Act

CFR - Code of Federal Regulations

Common Facilities - Recycled water distribution facilities connected with more than one distributor.

CWC - California Water Code

CSJC - City of San Juan Capistrano

CVWD - Capistrano Valley Water District

CSC - City of San Clemente

ETWD - El Toro Water District

Executive Officer - Executive Officer of the California Regional Water Quality Control Board, San Diego Region

IRWD - Irvine Ranch Water District

LAWD - Los Alisos Water District

MNWD - Moulton Niguel Water District

NPDES - National Pollutant Discharge Elimination System

Recycled Water - Water which, as a result of treatment of waste, is suitable for a direct beneficial use or a controlled use that would not otherwise occur (Water Code Section 13050(n)).

Recycled Water Agency - means the public water system, or a publicly owned or operated recycled water system, proposing to deliver recycled water to a facility.

Recycled Water Producer - Any person, agency, business, or organization treating wastewater so that it is suitable for direct beneficial use or controlled use that would not otherwise occur and allowing such uses to occur.

Recycled Water Purveyance Requirements - Requirements issued by a Regional Water Board to regulate the producer, distributor, and/or user of recycled water.

RWQCB - Regional Water Quality Control Board

SMWD - Santa Margarita Water District

SCWD - South Coast Water District

SERRA - South East Regional Reclamation Authority

SOCRA - South Orange County Reclamation Authority

TDS - Total Dissolved Solids

TCWD - Trabuco Canyon Water District

User - Any person, group, firm, partnership, corporation, association, agency, business, or organization receiving recycled water directly or indirectly from a producer and thereafter either distributing the water to others or applying it to a beneficial use. (There may be more than one user for any single producer.)

WDR - Waste Discharge Requirements

12. Under California Water Code (CWC) Section 13263 the San Diego RWQCB shall prescribe waste discharge requirements for any discharge of waste within the region which could affect water quality in the region. Waste discharge requirements must ensure compliance with applicable water quality control plans and prevent pollution, contamination, and nuisance. Under Water Code Section 13523 the San Diego RWQCB must, if necessary to protect the 'public health, safety and welfare,' prescribe water reclamation requirements for water which is used or proposed to be used as recycled water. In lieu of issuing waste discharge requirements for each user of recycled water, the Regional Board, pursuant to CWC Section 13523.1, may establish master water reclamation requirements which are issued to a producer or distributor, or both. The joint powers agreement creating SOCRA authorizes SOCRA to receive a single order, on behalf of its member agencies, containing both waste discharge requirements and water recycling requirements governing the production, distribution, use, and discharge of recycled water within the SOCRA jurisdictional area.
13. The Regional Board, acting in accord with Section 13244 of the California Water Code, adopted the Water Quality Control Plan for the San Diego Basin (9), (hereinafter Basin Plan) on September 8, 1994. The Basin Plan was subsequently approved by the State Water Resources Control Board (SWRCB) on December 13, 1994. Subsequent revisions to the Basin Plan have also been adopted by the Regional Board and approved by the SWRCB. The Basin Plan contains beneficial uses and water quality objectives. The requirements of this Order are consistent with the Basin Plan.

Tables 7 and 8 show the various hydrologic areas within SOCRA and their designated beneficial uses for ground and surface waters, respectively.

Table 7
EXISTING AND POTENTIAL BENEFICIAL USES OF GROUND WATERS

HYDROLOGIC AREAS AND SUBAREAS	FOOT- NOTES	M U N	A G R	I N D	P R O C	G W R	R E C I
1.10 Laguna HA							
1.11 San Joaquin Hills HSA	A	•	•				
1.12 Laguna Beach HSA	A	•	•				
1.13 Aliso HSA	B	•	•				
1.14 Dana Point HSA	A	+	•				
1.20 Mission Viejo HA							
1.21 Oso HSA		•	•	•			
1.22 Upper Trabuco HSA		•	•	•			
1.23 Middle Trabuco HSA		•	•	•			
1.24 Gobernadora HSA		•	•	•			
1.25 Upper San Juan HSA		•	•	•			
1.26 Middle San Juan HSA		•	•	•			
1.27 Lower San Juan HSA	C	•	•	•			
1.28 Ortega HSA		•	•	•			
1.30 San Clemente HA							
- 1.31 Prima Deshecha HSA	B	•	•				
1.32 Segunda Deshecha		+					
1.40 San Mateo Canyon HA	B	•	•	•			

HSA = Hydrologic Subarea

HA = Hydrologic Area

Definitions: • = Existing Beneficial Use per the 1994 Water Quality Control Plan

+ = Excepted from MUN

Footnotes:

- A These beneficial uses do not apply to all lands on the coastal side of the inland boundary of the right-of-way of Pacific Coast Highway 1, and this area is excepted from the source-of drinking water policy. The beneficial uses for the remainder of HA 1.10 are as shown.
- B These beneficial uses do not apply westerly of the easterly boundary of the of the right-of-way of Interstate Highway 5 and this area is excepted from sources of drinking water policy. The beneficial uses for the remainder of the hydrologic area are as shown.
- C These beneficial uses do not apply to all lands on the coastal side of the inland boundary of the right-of-way of Pacific Coast Highway 1 west of the San Juan Creek channel and this area is excepted from the sources of drinking water policy. The beneficial uses for the remainder of HA 1.20 are as shown.

Table 8

EXISTING AND POTENTIAL BENEFICIAL USES OF INLAND SURFACE WATERS

HYDROLOGIC	BENEFICIAL USE																			
	M U N	A G R	I N D	P R O C	G W R	F R S H	N A V	P O W	R E C I	R E C 2	C O M M	W A R M	C O L D	S A L	W I L D	R A R E	M A R	M I G R	S P W N	S H E L
1.10 Laguna HA																				
1.11 San Joaquin HSA	+	*							o	*		*			*		-			
1.12 Laguna Beach HSA	+	*							o	*		*			*					
1.13 Aliso HSA	+	*							o	*		*			*					
1.14 Dana Point HSA	+	*							o	*		*			*					
1.20 San Juan HA																				
1.21 Oso HSA	+	*	*						*	*		*	*		*					
1.22 Upper Trabuco HSA	+	*	*						*	*		*	*		*					
1.23 Middle Trabuco HSA	+	*	*						*	*		*	*		*					
1.24 Gobernadora HSA	+	*	*						*	*		*	*		*					
1.25 Upper San Juan HSA	+	*	*						*	*		*	*		*					
1.26 Middle San Juan HSA	+	*	*						*	*		*	*		*					
1.27 Lower San Juan HSA	+	*	*						*	*		*	*		*					
1.28 Ortega HSA	+	*	*						*	*		*	*		*					
1.30 San Clemente HA																				
1.31 Prima Deshecha HSA	+	*							o	*		*			*					
1.32 Segunda Deshecha HSA	+	*							o	*		*			*					
1.40 San Mateo HA	+								o	*		*			*	*				

HA = Hydrologic Area
HSA = Hydrologic Subarea

Definitions:

- * = Existing Beneficial Use per the Comprehensive Water Quality Control Plan 1994
- o = Potential Beneficial Use per the Comprehensive Water Quality Control Plan 1994
- +
- = Excepted from MUN
- * The Rare beneficial use applies only to portions of San Mateo Creek and to Wildhorse Canyon only.

Water quality objectives for ground and surface waters are shown in Tables 9 and 10, respectively.

Table 9

WATER QUALITY OBJECTIVES FOR GROUNDWATER

(Concentrations not to be exceeded more than 10% of the time during any one year period)															
(mg/L or as noted)															
HYDROLOGIC AREAS AND SUBAREAS	FOOT- NOTE	TDS	CL	SO ₄	% N a	N 0 3	Fe	Mn	M B A S	B	O D O R	TURB NTU	COL- OR UNITS	F	
1.10 Laguna HA															
1.11 San Joaquin Hill HSA		1200	400	500	60	10	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.12 Laguna Beach HSA		1200	400	500	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.13 Aliso HSA		1200	400	500	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.14 Dana Point HSA		1200	400	500	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.20 Mission Viejo HA															
1.21 Oso HSA		1200	400	500	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.22 Upper Trabuco HSA		500	250	250	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.23 Middle Trabuco HSA		750	375	375	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.24 Gobernadora HSA		1200	400	500	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.25 Upper San Juan HSA		500	250	250	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.26 Middle San Juan HSA		750	375	375	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.27 Lower San Juan HSA		1200	400	500	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.28 Ortega HSA		1100	375	450	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.30 San Clemente HA															
1.31 Prima Deshecha HSA		1200	400	500	60	10	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.32 Segunda Deshecha HSA		1200	400	500	60	10	0.3	0.05	0.5	0.75	NONE	5	15	1.0	
1.40 San Mateo Canyon HA	A,B	500	250	250	60	45	0.3	0.05	0.5	0.75	NONE	5	15	1.0	

HA = Hydrologic Area
HSA = Hydrologic Subarea

Source: Water Quality objectives per the Comprehensive Water Quality Control Plan 1994.

Footnotes:

- A The water quality objectives do not apply west of the easterly boundary of Interstate Highway 5. The objectives for the remainder of the Hydrologic Area (Subarea) are as shown.
- B Detailed salt balance studies are recommended for this area to determine limiting mineral concentration levels for discharge. On the basis on existing data, the tabulated objectives would probably be maintained in most areas. Upon completion of the salt balance studies, significant water quality objective revisions may be necessary. In the interim period of time, projects of ground water recharge with water quality inferior to the tabulated numerical values may be permitted following individual review and approval by the Regional Board if such projects do not degrade existing ground water quality to the aquifers affected by the recharge.

Table 10

WATER QUALITY OBJECTIVES FOR INLAND SURFACE WATERS

(Concentrations not to be exceeded more than 10% of the time during any one year period)													
(mg/L or as noted)													
HYDROLOGIC AREAS AND SUBAREAS	TDS	CL	SO ₄	% Na	N & P	Fe	Mn	M B A S	B	O D O R	TURB NTU	COLOR UNITS	F
1.10 Laguna HA													
1.11 San Joaquin Hills HSA	1000	400	500	60	A	0.3	0.05	0.5	0.75	NONE	20	20	1.0
1.12 Laguna Beach HSA	1000	400	500	60	A	0.3	0.05	0.5	0.75	NONE	20	20	1.0
1.13 Aliso HSA	1000	400	500	60	A	0.3	0.05	0.5	0.75	NONE	20	20	1.0
1.14 Dana Point HSA	1000	400	500	60	A	0.3	0.05	0.5	0.75	NONE	20	20	1.0
1.20 Mission Viejo HA	500	250	250	60	A	0.3	0.05	0.5	0.75	NONE	20	20	1.0
1.30 San Clemente HA	500	250	250	60	A	0.3	0.05	0.5	0.75	NONE	20	20	1.0
1.40 San Mateo Canyon HA	500	250	250	60	A	0.3	0.05	0.5	0.75	NONE	20	20	1.0

HA = Hydrologic Area
HSA = Hydrologic Subarea

Source: Water Quality objectives per the Comprehensive Water Quality Control Plan 1994.

Footnotes:

- A Concentrations of nitrogen and phosphorus, by themselves or in combination with other nutrients, shall be maintained at levels below those which stimulate algae and emergent plant growth. Threshold total Phosphorus (P) concentrations shall not exceed 0.05 mg/L in any stream at the point where it enters any standing body of water, nor 0.025 mg/L in any standing body of water. A desired goal in order to prevent plant nuisances in streams and other flowing waters appears to be 0.1 mg/L total P. These values are not to be exceeded more than 10% of the time, unless studies of the specific water body in question clearly show that water quality objective changes are permissible and changes are approved by the Regional Board. Analogous threshold values have not been set for nitrogen compounds; however, natural ratios of nitrogen to phosphorus are to be determined by surveillance and monitoring and upheld. If data is lacking, a ratio of N:P = 10:1 shall be used. Note - Certain exceptions to the above water quality objectives are described in Chapter 4 (1994 Comprehensive Water Quality Control Plan) in the Sections titled Discharges to Coastal Lagoons from Pilot Reclamation Projects and Discharges to Surface Waters.

South Orange County Reclamation Authority

14. Waste Discharge Requirements may be waived for short term usage of recycled water for soil compaction, dust control and other construction purposes as authorized by the Basin Plan, described in Chapter 4, Table 4-4.
15. Wastewater recycling and reuse projects that will occur in the areas under the terms and conditions of this Order will:
 - have maximum benefit to the people of the State, because in the absence of recycled wastewater, imported potable water would be used for irrigation of the recycled water use areas permitted in this Order;
 - not unreasonably affect the beneficial uses of groundwater in the hydrologic areas in which the use of recycled water has been permitted; and
 - not cause the groundwater quality objectives of the hydrologic subareas as currently designated to be exceeded.
16. The Basin Plan has established a policy for the implementation of ground water quality objectives for discharges of recycled water. The policy encourages the use of adequately treated recycled water for irrigation or groundwater recharge in basins where reuse is clearly beneficial.
17. Ocean disposal is permitted via ocean outfall permits for SERRA (Order No. 95-01) and for AWMA (Order No. 95-107) which are not within the authority of SOCRA. This Order does not replace the existing NPDES permits for the treatment and disposal of secondary effluent to the ocean.
18. On September 10, 1992, South Orange County Reclamation Authority approved a Negative Declaration for the use of recycled water within the SOCRA jurisdictional area in accordance with the California Environmental Quality Act (Public Resources Code Section 21000, et seq.), after a public hearing was held. The project as approved by SOCRA will not have a significant effect on water quality.
19. The Final Environmental Impact Report for the SMWD Oso Creek Water Reclamation Plant (WRP) identified a significant water resource related effect of the project: possible degradation of ground and surface water quality in the Oso Creek basin. The Oso Creek Barrier was constructed to assure that possible adverse impacts resulting from the SMWD Oso Creek WRP would be confined to the Oso Creek Basin and that the San Juan Basin would not be degraded. Flows captured at the Oso Creek Barrier are pumped to the Oso Creek WRP and mixed with the plant effluent for discharge to Upper Oso Reservoir.

20. The Basin Plan established the following water quality objectives for waters designated for use as domestic or municipal supply:

CHEMICAL	MAXIMUM CONTAMINANT LEVEL, MG/L
Aluminum	1.
Antimony	0.006
Arsenic	0.05
Asbestos	7 MFL *
Barium	1.
Beryllium	0.004
Cadmium	0.005
Chromium	0.05
Cyanide	0.2
Mercury	0.002
Nickel	0.1
Nitrate (as NO ₃)	45.
Nitrate + Nitrite (sum as nitrogen)	10.
Nitrite (as nitrogen)	1.
Selenium	0.05
Thallium	0.002
* MFL = million fibers per liter, MCL for fibers exceeding 10 micro-meters in length	

21. This Order prescribes waste discharge and water recycling requirements for SOCRA governing the production, distribution, use and discharge of recycled water within the SOCRA service area which the Regional Board has determined are necessary to protect the public health, safety and welfare pursuant to California Water Code, Division 7, Chapter 7, Sections 13500-13550 ("Water Reclamation Law"). This Order requires that SOCRA ensure compliance by its member agencies and recycled water users within its service area with the terms

and conditions of this Order. This Order requires that SOCRA establish and enforce minimum rules and regulations governing the production, distribution, and use of recycled water within the SOCRA service area, which ensure compliance with this Order.

22. The San Diego RWQCB, in establishing the requirements contained herein, considered factors including, but not limited to, the following:
 - (a) Beneficial uses to be protected and water quality objectives reasonably required for that purpose;
 - (b) Other waste discharges;
 - (c) The need to prevent nuisance;
 - (d) Past, present, and potential future beneficial uses of the hydrologic units being considered;
 - (e) Environmental conditions of the hydrologic units;
 - (f) Water quality conditions that could reasonably be achieved through the coordinated control of all factors which affect water quality in the area;
 - (g) Economic considerations;
 - (h) The need for developing housing within the region;
 - (i) The need for reliable water supplies into the area; and
 - (j) The need to develop and use recycled water.
23. The Regional Board has considered all water resource related environmental factors associated with the proposed discharge of waste.
24. The Regional Board has notified SOCRA, all SOCRA members and all known interested parties of the intent to prescribe waste discharge requirements for the proposed discharge.
25. The Regional Board in a public meeting heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED, that SOCRA shall, in order to meet the provisions contained in Division 7 of the California Water Code and Regulations adopted thereunder, comply and ensure compliance by its member agencies and their users with the following requirements for the discharge and purveyance of recycled water within the SOCRA service area:

A. DISCHARGE SPECIFICATIONS

1. The discharge for landscape irrigation and other land disposal projects of an effluent containing pollutants in excess of the following limitations is prohibited:

Table A-1

EFFLUENT LIMITATIONS FOR RECYCLED WATER COMMON TO ALL FACILITIES WITHIN THE SOCRA AREA

CONSTITUENT	12 month ¹ Avg. (mg/L)	Daily ² Maximum (mg/L)
Biochemical Oxygen Demand (BOD ₅ @ 20°C)	30	45
Total Suspended Solids	30	45
pH (within limits shown at all times)	6.0 - 9.0	6.0 - 9.0
Total Dissolved Solids	1000	1100
Chloride	375	400
Sulfate	400	450
% Sodium	60	65
Nitrate	45	50
Iron	0.3	0.4
Manganese	0.05	0.06
MBAS	0.5	0.6
Boron	0.67	0.75
Odor	NONE	NONE
Aluminum	1.0	
Arsenic	0.05	
Antimony	0.006	
Barium	1.0	
Beryllium	0.004	
Cadmium	0.005	
Chromium	0.05	
Cyanide	0.2	
Mercury	0.002	
Nickel	0.1	
Selenium	0.05	
Thallium	0.002	

- 1 The 12 month average effluent limitation shall apply to the arithmetic mean of the results of all samples collected during any 12 month consecutive calendar period.
- 2 The daily maximum effluent limitation shall apply to the results of a single composite or grab sample.

2. The 30-day average flow from each facility shall not exceed the Title 22 certified design capacity for that facility. The cumulative discharge from all SOCRA agency treatment facilities shall not exceed **52.36 MGD**.
3. The total annual recycled water use shall not exceed the volume specified in Table 6 of this Order.
4. Total coliform concentration of recycled water produced shall not exceed a MPN (most probable number) of 2.2 per 100 mL, utilizing the bacteriological results of the last seven days for which analyses have been completed and shall not exceed a MPN of 23 per 100 mL in more than one sample in any 30-day period.
5. Turbidity concentration of recycled water produced shall not exceed a daily average value of 2 NTU (Nephelometric turbidity unit), and shall not exceed 5 NTU more than 5% of the time during a 24-hour period and shall not exceed 10 NTU at any time.
6. Disinfection must comply with all requirements of the California Administrative Code, Title 22, Division 4. When chlorine is the disinfectant, disinfection of recycled water shall be accomplished by a chlorine disinfection process that provides a CT (chlorine concentration times modal contact time) value of not less than 450 mg-min/liter at all times with a modal chlorine contact time of at least 90 minutes based on peak design flow. Modal Contact Time is defined as the amount of time elapsed between the time that a tracer, such as salt or dye, is injected into the influent at the entrance to a chamber and the time that the highest concentration of the tracer is observed in the effluent from the chamber.
7. Discharges of recycled water shall not cause a violation of any Prohibition contained in the Water Quality Control Plan.
8. Disposal of captured Oso Creek Barrier flows shall be consistent with Finding 19 of this Order unless an alternative method of disposal is approved by the Executive Officer.

B. RECYCLED WATER PURVEYANCE REQUIREMENTS

- 1- SOCRA shall ensure that all recycled water producers within its jurisdictional area shall establish new, or have existing, **Rules and Regulations for Recycled Water Users** governing the design and construction of recycled water use facilities and the use of recycled water. SOCRA shall ensure that the rules and regulations are implemented and enforced with respect to the recycled water users. SOCRA shall certify, within 30 days of the

adoption of this Order, that SOCRA's and SOCRA's member agencies' Rules and Regulations for Recycled Water Users are consistent with Attachment 1 to this Order.

2. SOCRA shall certify, within 30 days of the adoption of this Order, that all recycled water distributors have developed and submitted to SOCRA a program to conduct compliance inspections of recycled water use sites to determine the status of compliance with the Rules and Regulations for Recycled Water Users.
3. SOCRA shall certify, within 30 days of the adoption of this Order, that the following information has been developed and is available to the State and County Health Departments for all current recycled water use areas:
 - a) The number, location, and type of facilities within the use area proposing to use domestic and recycled water. "Facility" means any type of building or structure, or defined area of specific public use that utilizes or proposes to utilize a dual plumbed system.
 - b) The average number of persons estimated to be served at each use area on a daily basis.
 - c) The specific boundaries of the proposed use site area including a map showing the location of each facility, drinking water fountain and impoundment to be served.
 - d) The person or persons responsible for operation of the recycled water system at each use area.
 - e) The specific use to be made of the recycled water at each use area.
 - f) The methods to be used by the recycled water agency to assure that the installation and operation of the recycled water system will not result in cross connections between the recycled water piping system and the potable water piping system. This shall include a description of pressure, dye or other test methods to be used to test the system.
 - g) Plans and specifications shall include the following and shall be submitted upon request to the State and County Health Departments:
 - 1) Proposed piping system to be used,
 - 2) Pipe locations of both the recycled and potable systems,

- 3) Type and location of the outlets and plumbing fixtures that will be accessible to the public,
 - 4) The methods and devices to be used to prevent backflow of recycled water into the public water system,
 - 5) Plan notes relating to recycled water specific installation and use requirements.
4. Prior to providing recycled water to a new use site, SOCRA shall ensure that the recycled water agency has done the following:
 - a) Submitted for review and approval a report that either certifies that the new use site conforms with what is described in the master plan or documents that all criteria described in *Recycled Water Purveyance Requirements B.3* has been submitted to and approved by the appropriate regulatory agency. Information submitted as a supplement to the master plan shall document compliance with any criteria, as described by *Recycled Water Purveyance Requirements B.3*, not met through submittal of the master plan.
 - b) Submitted for review and approval the information required by Monitoring and Reporting Program No. 97-52 E.2a, Recycled Water Users Summary Report, for the new use site.
5. SOCRA, or the appropriate member agency, shall do the following for all use sites:
 - a) Enforce recycled water rules and regulations,
 - b) Conduct recycled water use site compliance inspections in accordance with the program submitted in compliance with *Recycled Water Purveyance Requirements B.2* of this Order,
 - c) Notify the State Department of Health Services and the County of Orange Department of Environmental Health of any incidence of recycled water backflow into the potable water system as soon as possible, but in no case later than 24 hours of finding the incident,
 - d) Maintain a current list of all on-site recycled water supervisors.
6. Recycled water shall not be supplied to parties who use, transport, or store such water in a manner which causes a pollution, contamination or nuisance, as defined by Section 13050 of the California Water Code.

7. SOCRA shall ensure that users of recycled water designate a recycled water supervisor responsible for the recycled water system at each use area under the user's control. Recycled water supervisors should be responsible for the installation, operation, and maintenance of the irrigation system, as well as enforcement of rules and regulations, prevention of potential hazards, and maintenance of the distribution system plans in "as built" form.
8. Prior to using recycled water or supplying recycled water for use by other parties in any manner or in any area other than as described in the Findings of this Order, SOCRA shall obtain an amendment to this Order.
9. All use of recycled water shall be in accord with the operational plan for the use of recycled water submitted by the user and approved by SOCRA.

C. FACILITY DESIGN AND OPERATION SPECIFICATIONS

1. ENFORCEMENT

The Regional Board may initiate enforcement action against SOCRA or any member agency, which may result in the termination of the recycled water supply, if any person uses, transports, or stores such water in a manner which creates, or threatens to create conditions of pollution, contamination, or nuisance, as defined in Water Code Section 13050.

2. PROPER OPERATION

SOCRA shall ensure that all facilities and systems of treatment, distribution, and control (and related appurtenances) which are installed or used to achieve compliance with conditions of this Order are at all times properly operated and maintained. Proper operation and maintenance includes effective performance, adequate funding, adequate operator staffing and training, and adequate laboratory and process controls, including appropriate quality assurance procedures. This provision requires the operation of backup and auxiliary facilities or similar systems only when necessary to achieve compliance with the conditions of this Order.

3. OPERATION MANUAL

A copy of the facility operations manual shall be maintained at each facility and shall be available to operation personnel and Regional Board staff at all times. The following portions of the operations manual shall be posted at the treatment plant as a quick reference for treatment plant operators:

- a) Alarm set points for secondary turbidity, tertiary turbidity and chlorine residual,
- b) Levels at which flow will be diverted for secondary turbidity, tertiary turbidity and chlorine residual.
- c) When to divert flow for high daily and weekly median total coliform;
- d) When the authorities (DHS, EHD, Regional Board) will be notified of a diversion,
- e) Names and numbers of those authorities to be notified in case of a diversion,
- f) Frequency of calibration for turbidimeters and chlorine residual analyzers.

4. OPERATORS' CERTIFICATION

The wastewater treatment facilities shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to Chapter 3, Subchapter 14, Title 23 of the California Code of Regulations.

5. FLOOD AND RUNOFF PROTECTION

All water recycling facilities shall be protected against 100-year peak stream flows as defined by the Orange County flood control agency and shall also be protected from overland runoff and other impacts resulting from a 100-year frequency 24-hour storm. The Upper Oso Reservoir shall be operated to contain 100-year storm and seasonal runoff, prevent seepage into Oso Creek, capture low surface and subsurface flows, and prevent violation of surface and groundwater objectives.

6. MONITORING AND REPORTING

SOCRA and its member agencies shall comply with the attached Monitoring and Reporting Program No. 97-52, and future revisions thereto as specified by the Executive Officer. Monitoring results shall be reported to SOCRA at the intervals specified in Monitoring and Reporting Program No. 97-52. Any increase in the maximum waste discharge flow specified for each recycled water production facility in Table 5, and/or increase in the volume of use specified for each HSA in Table 6, other than increases requiring a new report of waste discharge, shall be included in SOCRA's summary data. SOCRA shall submit monitoring results and summary data quarterly to the San Diego RWQCB.

7. EFFLUENT STORAGE FACILITIES

Effluent storage facilities shall be designed, constructed, operated, and maintained so as to prevent surfacing of wastes on property not owned or controlled by the facility owner or operator. Surface runoff of any wastes onto property not owned or controlled by the owner or operator shall be prevented.

8. DESIGN, OPERATION AND RELIABILITY

The producers shall meet the design, operational, and reliability requirements of Articles 7, 8, 9 and 10 of the California Code of Regulations, Title 22, Division 4, Chapter 3.

D. BIOSOLIDS SPECIFICATIONS

1. Collected screenings, sludges, other solids removed from liquid wastes, and filter backwash shall be disposed in a manner approved by the Executive Officer.
2. Management of all solids and sludge must comply with all requirements of 40 CFR Parts 257, 258, 501 and 503; CWA Part 405(d), and Title 23, CCR, Chapter 15, including all monitoring, record keeping and reporting requirements. Since the State of California, hence the State and Regional Boards, has not been delegated the authority by the USEPA to implement the sludge program, enforcement of sludge requirements of CFR Part 503 is under USEPA's jurisdiction. Once sludge leaves a facility, it is subject to all applicable local, state and federal laws and regulations.
3. All solids and sludge must be disposed of in a municipal solid waste landfill, reused by land application or disposed of in a sludge-only landfill accordance with 40 CFR Parts 503 and 258, and Title 23 CCR Chapter 15. If the discharger desires to dispose of solids or sludge by a different method, a request for permit modification must be submitted to the USEPA and this Regional Board 180 days prior to the initiation of the alternative disposal.
4. Solids and sludge treatment, storage, and disposal or reuse shall not create a nuisance, such as objectionable odors or flies, and shall not result in groundwater contamination.
5. The solids and sludge treatment site and storage site shall have facilities adequate to divert surface water runoff from adjacent areas, to protect the boundaries of the site from erosion, and to prevent drainage from the

treatment and storage site. Adequate protection is defined as protection from at least a 100-year storm and protection from the highest possible tidal stage that may occur.

6. The discharge of sewage sludge and solids shall not cause waste material to be in a position where it is, or can be, conveyed from the treatment and storage sites and deposited in the waters of the state.
7. The discharger shall submit a copy of each of the annual reports required by 40 CFR 503 to this Regional Board Executive Officer at the same time those reports are submitted to USEPA. The discharger shall also submit an annual report of the quantity and disposition of sludge generated in the previous calendar year.

E. STANDARD PROVISIONS

1. DUTY TO COMPLY

SOCRA shall ensure that all member agencies and recycled water users comply with all applicable conditions of this Order. Any noncompliance with this Order is grounds for:

- enforcement action;
- termination, revocation and reissuance, or modification of the Order for which a noncompliance has occurred; or
- denial of a report of waste discharge in application for new or revised waste discharge requirements.

2. ENTRY AND INSPECTION

SOCRA shall ensure that the San Diego RWQCB, or an authorized representative, or any representative of the United States Environmental Protection Agency, the State Department of Health Service, or the County of Orange, Health Care Agency, Environmental Health, Water Quality Section, upon the presentation of proper credentials and other documents as may be required by law, may:

- enter upon any premises where a regulated facility or activity is located, or where records must be kept under the conditions of this Order;

- have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order;
- inspect at reasonable times any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order; and
- sample or monitor at reasonable times, for the purposes of assuring compliance with this Order or as otherwise authorized by the California Water Code, any substances or parameters at any location.

SOCRA shall provide written responses to requests by the San Diego RWQCB, the United States Environmental Protection Agency, the State Department of Health Service, and the County of Orange, Health Care Agency, Environmental Health, Water Quality Section, for any information relating to this Order or the San Diego RWQCB's responsibility under the Water Code.

3. CIVIL MONETARY REMEDIES

The California Water Code provides that any person who intentionally or negligently violates any waste discharge requirements issued, reissued, or amended by this Regional Board is subject to a civil monetary remedy of up to 20 dollars per gallon of waste discharged or, if a cleanup and abatement order is issued, up to 15,000 dollars per day of violation or some combination thereof.

4. PENALTIES FOR INVESTIGATION, MONITORING OR INSPECTION VIOLATIONS

The California Water Code provides that any person failing or refusing to furnish technical or monitoring program reports, as required under this Order, or falsifying any information provided in the monitoring reports is guilty of a misdemeanor and is subject to a civil liability of up to 5,000 dollars for each day in which the violation occurs.

5. ENDANGERMENT OF HEALTH AND ENVIRONMENT

SOCRA or the appropriate member agency, shall report any noncompliance which may endanger health or the environment. Any such information shall be provided orally to the San Diego RWQCB within 24 hours from the time SOCRA or the agency becomes aware of the circumstances. A written submission shall also be provided within five days of the time SOCRA becomes aware of the circumstances. The written submission shall contain a description of the noncompliance and its cause; the period

of noncompliance, including exact dates and times; and if the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent recurrence of the noncompliance. The San Diego RWQCB, or an authorized representative, may waive the written report on a case-by-case basis if the oral report has been received within 24 hours. The Executive Officer shall be informed by SOCRA of all such notifications and of the resultant actions taken. The following occurrence(s) must be reported to the San Diego RWQCB Executive Officer within 24 hours:

- Any bypass from any portion of the treatment facility.
- Any discharge of treated or untreated wastewater resulting from sewer line breaks, obstruction, surcharge, or any other circumstances.
- Any treatment plant upset which causes the effluent limitations of this Order to be exceeded.

6. PRIOR NOTICE OF BYPASS

If a need for a discharge bypass is known in advance, the producer or distributor shall submit prior notice to SOCRA and, if at all possible, such notice shall be submitted at least 10 days prior to the date of the bypass. Discharge bypasses must be reported to the San Diego RWQCB by SOCRA.

7. CORRECTIVE ACTION

SOCRA, or the appropriate member agency, shall take all reasonable steps to minimize or correct any adverse impact on the environment resulting from noncompliance with this Order, including such accelerated or additional monitoring as may be necessary to determine the nature and impact of the noncompliance.

8. TREATMENT FAILURE

In an enforcement action, it shall not be a defense for SOCRA, its member agencies or recycled water users that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with this Order. Upon reduction, loss, or failure of the treatment facility, the recycled water producer shall, to the extent necessary to maintain compliance with this Order, control production or all discharges, or both, until the facility is restored or an alternative method of treatment is provided. This

provision applies for example, when the primary source of power of the treatment facility fails, is reduced, or is lost.

9. HAZARDOUS RELEASES

Except for a discharge which is in compliance with these waste discharge requirements, any person who, without regard to intent or negligence, causes or permits any hazardous substance or sewage to be discharged in or on any waters of the State, shall as soon as (a) that person has knowledge of the discharge, (b) notification is possible, and (c) notification can be provided without substantially impeding cleanup or other emergency measures, immediately notify the Director of Environmental Health Division, County of Orange in accordance with California Health and Safety Code Section 5411.5 and the Office of Emergency Services of the discharge in accordance with the spill reporting provision of the State toxic disaster contingency plan adopted pursuant to Article 3.7 (commencing with Section 8574.7) of Chapter 7 of Division 1 of Title 2 of the Government Code, and immediately notify the State Board or the appropriate Regional Board of the discharge. This provision does not require reporting of any discharge of less than a reportable quantity as provided for under subdivisions (f) and (g) of Section 13271 of the Water Code unless the recycled water agency is in violation of a prohibition in the applicable Water Quality Control Plan.

10. PETROLEUM RELEASES

Except for a discharge which is in compliance with these waste discharge requirements, any person who without regard to intent or negligence, causes or permits any oil or petroleum product to be discharged in or on any waters of the State, or discharged or deposited where it is, or probably will be, discharged in or on any waters of the State, shall, as soon as (a) such person has knowledge of the discharge, (b) notification is possible, and (c) notification can be provided without substantially impeding cleanup or other emergency measures, immediately notify the Office of Emergency Services of the discharge in accordance with the spill reporting provision of the State oil spill contingency plan adopted pursuant to Article 3.5 (commencing with Section 8574.1) of Chapter 7 of Division 1 of Title 2 of the Government Code. This requirement does not require reporting of any discharge of less than 42 gallons unless the discharge is also required to be reported pursuant to Section 311 of the Clean Water Act or the discharge is in violation of a prohibition in the applicable Water Quality Control Plan.

11. PERMIT REPOSITORY

A copy of this Order shall be maintained and posted at a prominent location at or near each treatment and disposal facility subject to this Order, and shall be available to operating personnel at all times.

12. RETENTION OF RECORDS

SOCRA, or the appropriate member agency, shall retain records of all monitoring information, including all calibration and maintenance records, copies of all reports required by this Order, and records of all data used to complete the application for this Order. Records shall be maintained for a minimum of five years from the date of the sample, measurement, report, or application. This period may be extended during the course of any unresolved litigation regarding this discharge or when requested by the Regional Board Executive Officer.

13. GENERAL REPORTING REQUIREMENT

SOCRA member agencies shall furnish to SOCRA, to be forwarded to the Executive Officer, within a reasonable time, any information which the Executive Officer may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order. SOCRA member agencies shall also furnish to the Executive Officer through SOCRA, upon request, copies of records required to be kept by this Order.

14. PERMIT REVISION

This Order may be modified, revoked and reissued, or terminated for cause including, but not limited to:

- violation of any terms or conditions of this Order;
- obtaining this Order by misrepresentation or failure to disclose fully all relevant facts; or
- a change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge.

The filing of a request by SOCRA for the modification, revocation and reissuance, or termination of this Order, or notification of planned changes or anticipated noncompliance does not stay any condition of this Order.

15. CHANGE IN DISCHARGE

SOCRA shall file a new Report of Waste Discharge at least 120 days prior to the following:

- Addition of a major industrial waste discharge to a discharge of essentially domestic sewage, or the addition of a new process or product by an industrial facility resulting in a change in the character of the wastes. A change in the character of the wastes includes the addition of pollutants not previously present in the waste stream, or an increase of 25% or greater of the flow volume or concentration of any previously permitted pollutant.
- Significant change in the treatment or disposal method (e.g., change in the method of treatment which would significantly alter the nature of the waste.)
- Change in the disposal area for any waste product other than recycled water from that described in the Findings of this Order.
- Increase in proposed volume of use in any hydrologic area beyond that specified in Table 6 of this Order.
- Increase in proposed waste discharge flow for any facility beyond that specified in Table 5 of this Order.
- Other circumstances which result in a material change in character, amount, or location of the waste discharge.
- Any planned change in any regulated facility or activity which may result in noncompliance with this Order.
- Any other significant planned physical alterations or additions to the permitted facility.

16. CHANGE IN OWNERSHIP

This Order is not transferrable to any person except after notice to the Executive Officer. SOCRA shall submit this notice in writing at least 30 days in advance of any proposed transfer. The notice must include a written agreement between SOCRA and new responsible party containing a specific date for the transfer of this Order's responsibility and coverage between SOCRA and the new responsible party. This agreement shall include an acknowledgement that SOCRA is liable for violations up to the transfer date and that the new responsible party is liable from the transfer

date on. The Regional Board may require modification or revocation and reissuance of this Order to change the name of the responsible party and incorporate such other requirements as may be necessary under the California Water Code.

17. INCOMPLETE REPORTS

Where SOCRA becomes aware that it failed to submit any relevant facts in a Report of Waste Discharge or submitted incorrect information in a Report of Waste Discharge, or in any report to the San Diego RWQCB, it shall promptly submit such facts or information to the San Diego RWQCB.

18. REPORT DECLARATION

All applications, reports, or information submitted to the Executive Officer shall be signed and certified as follows:

- a) The Report of Waste Discharge shall be signed as follows:
 - 1) For a corporation - by a principal executive officer of at least the level of vice-president.
 - 2) For a partnership or sole proprietorship - by a general partner or the proprietor, respectively.
 - 3) For a municipality, state, federal or other public agency - by either a principal executive officer or ranking elected official.
- b) All other reports required by this Order and other information required by the Executive Officer shall be signed by a person designated in paragraph (a) of this provision, or by a duly authorized representative of that person. An individual is a duly authorized representative only if all of the following are true:
 - 1) The authorization is made in writing by a person described in paragraph (a) of this provision,
 - 2) The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity, and
 - 3) The written authorization is submitted to the Executive Officer.

- c) Any person signing a document under this Section shall make the following certification,

"I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment."

19. REGIONAL BOARD ADDRESS

The member agencies shall submit reports required under this Order, or other information required by the Executive Officer, to SOCRA, whose current address is:

South Orange County Reclamation Authority
30290 Rancho Viejo Road
San Juan Capistrano, CA 92675

All member agencies shall be informed of any future change of address by SOCRA.

SOCRA shall submit all reports required by this Order to the San Diego RWQCB, whose current address is:

Executive Officer
California Regional Water Quality Control Board
San Diego Region
9771 Clairemont Mesa Blvd, Suite A
San Diego, California 92124-1324

SOCRA shall be informed of any future change of address by the San Diego RWQCB.

F. NOTIFICATIONS

1. VESTED RIGHTS

This Order does not convey any property rights of any sort or any exclusive privileges. The requirements prescribed herein do not authorize the commission of any act causing injury to persons or property, nor protect the recycled water agency from liability under federal, state or local

laws, nor create a vested right for the recycled water agency to continue the waste discharge.

2. U.S. EPA REVIEW

These requirements have not been officially reviewed by the United States Environmental Protection Agency and are not issued pursuant to Section 402 of the Clean Water Act.

3. SEVERABILITY

The provisions of this Order are severable, and if any provision of this Order, or the application of any provision of this Order to any circumstance, is held invalid, the application of such provision to other circumstances, and the remainder of this Order, shall not be affected thereby.

4. FACILITY MODIFICATIONS

For any future treatment plant modifications to increase recycled water capacity production, the facility owner or operator shall file an engineering report conforming to Section 60323, Article 7 of the California Code of Regulations, Title 22, Division 4, Chapter 3. The engineering report shall be submitted to the State Department of Health Services, County Department of Health Services, San Diego RWQCB Executive Officer, and SOCRA. Increased volumes of recycled water, due to plant modifications, shall not be purveyed until the engineering report is approved by the Executive Officer.

5. SECTION 311 CLEAN WATER ACT

- Nothing in this Order shall be construed to preclude the institution of any legal action or relieve SOCRA from any responsibilities, liabilities, or penalties to which SOCRA is, or may be, subject to under Section 311 of the Clean Water Act.

6. SECTION 510 CLEAN WATER ACT

Nothing in this Order shall be construed to preclude the institution of any legal action or relieve SOCRA from any responsibilities, liabilities, or penalties established pursuant to any applicable State law or regulation under authority preserved by Section 510 of the Clean Water Act.

7. OSO CREEK BARRIER

Santa Margarita Water District shall report the volume of groundwater and surface water captured at the Upper Oso Creek Barrier. The method of disposal of these waters shall be consistent with Finding 19 of this Order. The volume of captured flows shall be reported monthly to SOCRA.

8. EFFECTIVE DATE

This Order becomes effective on the date of adoption by the San Diego RWQCB.

I, John H. Robertus, Executive Officer, do hereby certify the foregoing is a full, true and correct copy of an Order adopted by the California Regional Water Quality Control Board, San Diego Region on October 15, 1997.

A handwritten signature in dark ink, appearing to read "John H. Robertus", is written over a horizontal line.

**ATTACHMENT NO. 1
TO
ORDER NO. 97-52**

SOCRA INSTITUTIONAL BOUNDARIES

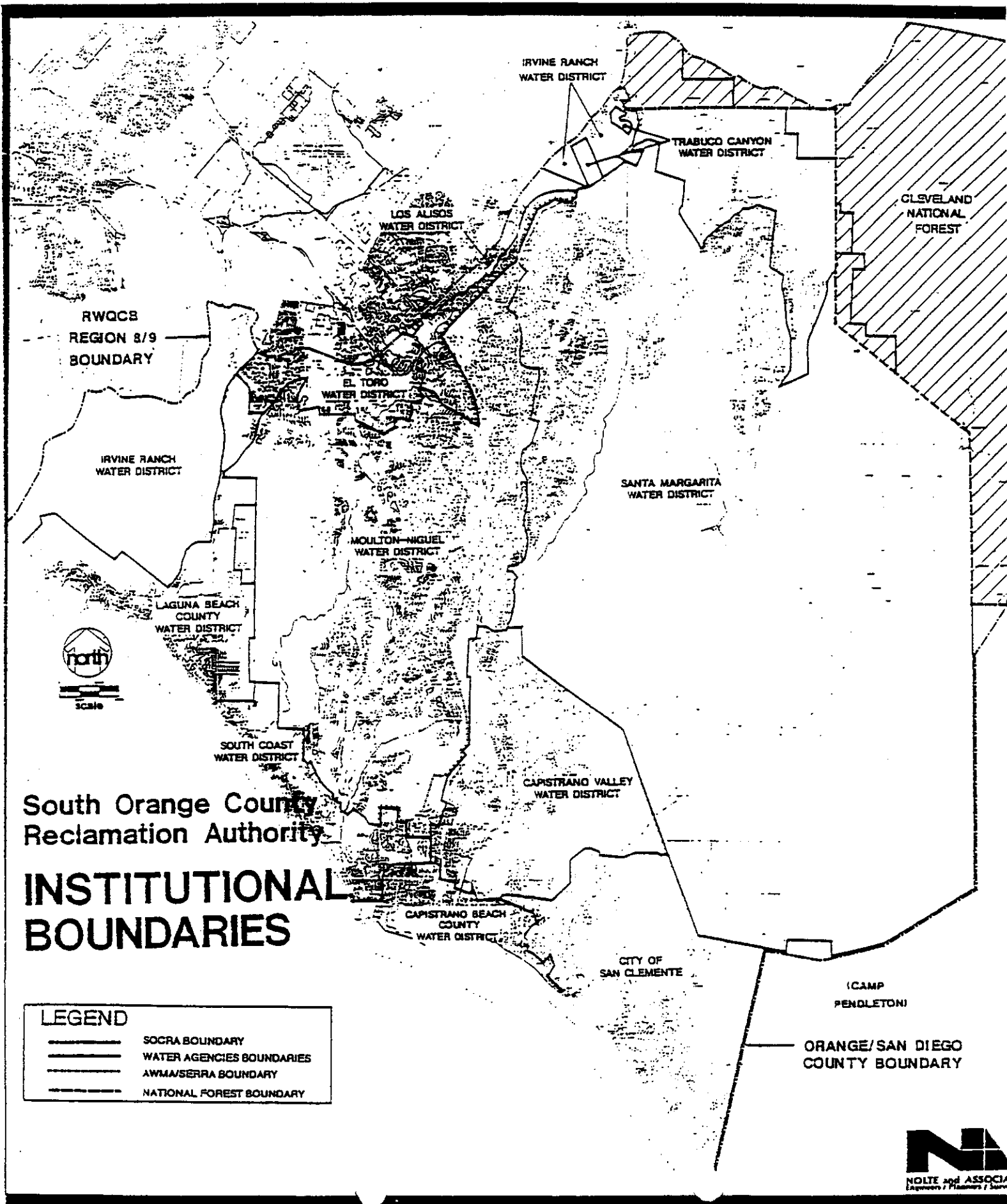


Figure 1

RULES AND REGULATIONS FOR RECYCLED WATER USE PROJECTS

Pursuant to California Water Code (CC) Section 13523.1(b)(3), this Order requires the recycled water agency to establish and to enforce rules and regulations governing the design, construction and use of recycled water distribution and disposal systems by its customers. The rules and regulations shall be consistent with the with the following criteria:

- a) Title 22, Division 4, Chapter 3, *Wastewater Reclamation Criteria*;
- b) Title 17, Division 1, Chapter 5, Group 4, Article 1 & 2, of the California Code of Regulations;
- c) Any measures that are deemed necessary for protection of public health, such as the American Water Works Association (AWWA) California/Nevada Section, *Guidelines for the Distribution of Non-Potable Water* or other measures that are deemed necessary by the State DOHS.

At a minimum, the rules and regulations shall notify the users that:

- 1. The use of recycled water shall not cause a pollution, contamination or nuisance, as defined by Section 13050 of the California Water Code.
- 2. The Recycled Water Agency, the Regional Board, the State and Local Health Department, or an authorized representative of these parties, upon presentation of proper credentials, shall have the right to enter upon the recycled water use site during reasonable hours, to verify that the user is complying with the Recycled Water Agency's rules and regulations.
- 3. The recycled water user shall provide written notification, in a timely manner, to the Recycled Water Agency of any material change or proposed change in the character of the use of recycled water.
- 4. Prior to the initiation of recycled water service, the recycled water user shall submit plans and specifications for recycled water distribution facilities to the Recycled Water Agency.
- 5. The recycled water user shall designate a recycled water supervisor who is responsible for the recycled water system at each use area under the user's control. Specific responsibilities of the recycled water supervisor include the proper installation, operation, and maintenance of the irrigation system; compliance of the project with the Recycled Water Agency's rules and regulations, prevention of potential hazards and preservation of the recycled water distribution system plans in "as built" form.

6. The Recycled Water Agency may terminate service to a recycled water user who uses, transports, or stores such water in violation of the Recycled Water Agency's rules and regulations.
7. All recycled water storage facilities owned and/or operated by recycled water users shall be protected against erosion, overland runoff, and other impacts resulting from a 100-year, 24 hour frequency storm unless the Regional Board Executive Officer approves relaxed storm protection measures for the facility.
8. All recycled water storage facilities owned and/or operated by recycled water users shall be protected against 100 - year frequency peak stream flows as defined by the Orange County flood control agency unless the Regional Board Executive Officer approves relaxed flood protection measures for the facility.
9. The Regional Board may initiate enforcement action against any recycled water user who discharges recycled water in violation of any applicable discharge requirement prescribed by the Regional Board or in a manner which creates or threatens to create conditions of pollution, contamination or nuisance, as defined in Water Code Section 13050.
10. A copy of the recycled water rules and regulations, irrigation system layout map, and a recycled water system operations manual shall be maintained at the use area. These documents shall be available to operating personnel at all times.
11. Irrigation with disinfected tertiary recycled water shall not take place within 50 feet of any domestic water supply well unless all of the following conditions have been met:
 - (a) A geological investigation demonstrates that an aquitard exists at the well between the uppermost aquifer being drawn from and the ground surface.
 - (b) The well contains an annular seal that extends from the surface into the aquitard.
 - (c) The well is housed to prevent any recycled water spray from coming into contact with the wellhead facilities.
 - (d) The ground surface immediately around the wellhead is contoured to allow surface water to drain away from the well.
 - (e) The owner of the well approves of the elimination of the buffer zone requirement.

12. Impoundment of disinfected tertiary recycled water shall not occur within 100 feet of any domestic water supply well.
13. Irrigation with, or impoundment of, disinfected secondary-2.2 or disinfected secondary -23 recycled water shall not take place within 100 feet of any domestic water supply well.
14. Irrigation with, or impoundment of, undisinfected secondary recycled water shall not take place within 150 feet of any domestic water supply well.
15. Recycled water facilities shall be operated in accordance with best management practices (BMP's) to prevent direct human consumption of recycled water and to minimize misting, ponding, and runoff. BMP's shall be implemented that will minimize both public contact and discharge onto areas not under customer control.
16. Irrigation with recycled water shall be during periods of minimal human use of the service area. Consideration shall be given to allow a maximum dry-out time before the irrigated area will be used by the public.
17. All drinking fountains located within the approved use area shall be protected by location and/or structure from contact with recycled water spray, mist, or runoff. Protection shall be by design, construction practice, or system operation.
18. Facilities that may be used by the public, including but not limited to eating surfaces and playground equipment and located within the approved use areas, shall be protected to the maximum extent possible by siting and/or structure from contact by irrigation with recycled water spray, mist, or runoff. Protection shall be by design, construction practice or system operation.
19. -Spray irrigation with recycled water, other than disinfected tertiary recycled water, shall not take place within 100 feet of the property line of a residence or a place where public exposure could be similar to that of a park, playground, or school yard.
20. All use areas where recycled water is used and that are accessible to the public shall be posted with conspicuous signs, in a size no less than 4 inches by 8 inches, that include the following wording: "RECYCLED WATER - DO NOT DRINK". The sign(s) shall be of a size easily readable by the public.
21. No physical connection shall be made or allowed to exist between any recycled water system and any separate system conveying potable water.

22. The recycled water piping system shall not include any hose bibs. Quick couplers that are different from that used on the potable water system may be used.
23. The public water supply shall not be used as a backup or supplemental source of water for a recycled water system unless the connection between the two systems is protected by an air gap separation which complies with the requirements of Sections 7602(a) and 7603(a) of Title 17 and the approval of the public water system has been obtained. If a "Swivel-ell" type connection is used it must be used in accordance with the provisions of the Department of Health Services Policy Memo 95-004. Approved backflow prevention devices shall be provided, installed, tested, and maintained by the recycled water user in accordance with the applicable provisions of Title 17, Division 1, Chapter 5, Group 4, Article 2.
24. No person other than the Recycled Water Agency shall deliver recycled water to a facility.
25. All recycled water piping and appurtenances in new installations and appurtenances in retrofit installations shall be colored purple or distinctively wrapped with purple tape in accordance with Chapter 7.9, section 4049.54 of the California Health and Safety Code.
26. Reuse site shut down tests and inspections shall be monitored by the County of Orange Environmental Health Division or the State Department of Health Services.
27. Customer complaints concerning recycled water use that may involve public illness shall be reported to the County of Orange Environmental Health Division and the State Department of Health Services, and to the Recycled Water Agency who shall maintain a log of all customer complaints regarding recycled water.
28. Any backflow prevention device installed to protect the public water system shall be inspected and maintained in accordance with Section 7605 of Title 17.

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
SAN DIEGO REGION**

MONITORING AND REPORTING PROGRAM NO. 97-52

**SOUTH ORANGE COUNTY RECLAMATION AUTHORITY,
ORANGE COUNTY**

A. MONITORING PROVISIONS

1. Samples and measurements taken as required herein shall be representative of the volume and nature of the monitored discharge. All samples shall be taken at the monitoring points specified in this permit, and unless otherwise specified, before the effluent joins or is diluted by any other waste stream, body of water, or substance. Monitoring points shall not be changed without notification to and the approval of the Regional Board Executive Officer.
2. Appropriate flow measurement devices and methods consistent with accepted scientific practices shall be selected and used to ensure the accuracy and reliability of measurements of the volume of monitored discharges. The devices shall be installed, calibrated, and maintained to ensure that the accuracy of the measurements are consistent with the accepted capability of that type of device. Devices selected shall be capable of measuring flows with a maximum deviation of less than ± 5 percent from true discharge rates throughout the range of expected discharge volumes. Guidance in selection, installation, calibration, and operation of acceptable flow measurement devices can be obtained from the following references:
 - "A Guide to Methods and Standards for the Measurement of Water Flow," U.S. Department of Commerce, National Bureau of Standards, NBS Special Publication 421, May 1975, 97 pp. (Available from the U.S. Government Printing Office, Washington D.C. 20402. Order by SD Catalog No. C13.10:421.)
 - "Water Measurement Manual," U.S. Department of Interior, Bureau of Reclamation, Second Edition, Revised Reprint, 1974, 327 pp. (Available from the U.S. Government Printing Office, Washington D.C. 20402. Order by Catalog No. 127, 19/2:W29/2, Stock No. S/N 24003-0027.)
 - "Flow Measurement in Open Channels and Closed Conduits," U.S. Department of Commerce, National Bureau of Standards, NBS Special Publication 484, October 1977, 982 pp. (Available in

paper copy or microfiche from National Technical Information Service (NTIS) Springfield, VA 22151. Order by NTIS No. PB-273-535/5ST.)

"NPDES Compliance Sampling Manual," U.S. Environmental Protection Agency, Office of Water Enforcement. Publication MCD-51, 1977, 140 pp. (Available from the General Services Administration (8FFS), Centralized Mailing Lists Services, Building 41, Denver Federal Center, Denver, CO 80225.)

3. Monitoring must be conducted according to United States Environmental Protection Agency test procedures approved under Title 40, Code of Federal Regulations (CFR), Part 136, "Guidelines Establishing Test Procedures for Analysis of Pollutants Under the Clean Water Act" as amended, unless other test procedures have been specified in this Order.
4. All analyses shall be performed in a laboratory certified to perform such analyses by the California Department of Health Services or a laboratory approved by the San Diego RWQCB Executive Officer.
5. Monitoring results must be reported on discharge monitoring report forms approved by the San Diego RWQCB Executive Officer.
6. If SOCRA, or any member agency, monitors any pollutants more frequently than required by this Order, using test procedures approved under 40 CFR, Part 136, or as specified in this Order, the results of this monitoring shall be included in the calculation and reporting of the data submitted in its monitoring report. The increased frequency of monitoring shall also be reported.
7. SOCRA, or the appropriate member agency, shall retain records of all monitoring information, including all calibration and maintenance records and copies or originals of all strip chart recordings for continuous monitoring instrumentation, copies of all reports required by this Order, and records of all data used to complete applications required by this Order. Records shall be maintained for a minimum of five years from the date of the sample, measurement, report, or application. This period may be extended during the course of any unresolved litigation regarding this discharge or when requested by SOCRA or the Executive Officer.
8. Records of monitoring information shall include:
 - the date, exact place, and time of sampling or measurements;

- the individual(s) who performed the sampling or measurements;
 - the date(s) analyses were performed;
 - the individual(s) who performed the analyses;
 - the analytical techniques or methods used; and
 - the results of such analyses.
9. All monitoring instruments and devices which are used by the producer or distributor to fulfill the prescribed monitoring program shall be properly maintained and calibrated as necessary to ensure their continued accuracy.
 10. The monitoring reports shall be signed by an authorized person as required by Reporting Requirements E.18 of this Order.
 11. A composite sample is defined as a combination of at least eight sample aliquots of at least 100 milliliters, collected at periodic intervals during the operating hours of a facility over a 24-hour period. For volatile pollutants, aliquots must be combined in the laboratory immediately before analysis. The composite must be flow proportional; either the time interval between each aliquot or the volume of each aliquot must be proportional to either the stream flow at the time of sampling or the total stream flow since the collection of the previous aliquot. Aliquots may be collected manually or automatically.
 12. A grab sample is an individual sample of at least 100 milliliters collected at a randomly selected time over a period not exceeding 15 minutes.
 13. Sampling and analysis shall, as a minimum, be conducted in accordance with Article 6 of California Code of Regulations, Title 22, Division 4, Chapter 3 (Reclamation Criteria).

B. EFFLUENT MONITORING

- Member agencies shall monitor the effluent from each of their treatment facilities described in Table 5 for the following constituents at the noted frequency:

Table B

DETERMINATION	UNIT	SAMPLE TYPE	SAMPLING FREQUENCY ¹	REPORTING FREQUENCY
Flowrate	MGD	Continuous	Continuous	Monthly
Biochemical Oxygen Demand (5-Day @ 20° C)	mg/L	Composite	1x/week	Monthly
Total Suspended Solids	mg/L	Composite	1x/week	Monthly
Volatile Suspended Solids	mg/L	Composite	1x/week	Monthly
pH	Unit	Composite	1x/week	Monthly
Total Dissolved Solids	mg/L	Composite	Quarterly	Quarterly
Chloride	mg/L	Composite	Quarterly	Quarterly
Adjusted Sodium Adsorption Ratio ²		Composite	Quarterly	Quarterly
Sulfate	mg/L	Composite	Quarterly	Quarterly
Iron	mg/L	Composite	Quarterly	Quarterly
Manganese	mg/L	Composite	Quarterly	Quarterly
Methylene Blue Active Substances	mg/L	Composite	Quarterly	Quarterly
Boron	mg/L	Composite	Quarterly	Quarterly
Aluminum	mg/L	Composite	Annual	Annual
Arsenic	mg/L	Composite	Annual	Annual
Barium	mg/L	Composite	Annual	Annual
Cadmium	mg/L	Composite	Annual	Annual
Chromium	mg/L	Composite	Annual	Annual
Copper	mg/L	Composite	Annual	Annual
Lead	mg/L	Composite	Annual	Annual
Mercury	mg/L	Composite	Annual	Annual
Selenium	mg/L	Composite	Annual	Annual
Silver	mg/L	Composite	Annual	Annual
Zinc	mg/L	Composite	Annual	Annual
Nitrate -Nitrogen	mg/L	Composite	Annual	Annual
Total Organic Carbon	mg/L	Composite	Quarterly	Quarterly
Chlorine Residual	mg/L	Continuous	Continuous	Monthly
Coliform	MPN/100 mL	Grab	Daily	Monthly
Turbidity	NTU	Continuous	Continuous	Monthly

- The sampling frequency shall be increased for any constituent that exceeds the limit specified by Discharge Specification B.1 of this Order. The increased frequencies shall be as follows:

1x/week to 3x/week
Quarterly to Monthly
Annually to Quarterly

The increased sampling frequency shall continue until compliance is achieved with the effluent limitations for three consecutive time frames (i.e. three weeks, three months or 3 quarters).

- 2 The adjusted sodium adsorption ratio is calculated as follows:

Adjusted Sodium Adsorption Ratio (Adj. SAR):
$$\frac{\text{Na}}{\sqrt{(\text{Ca}_x + \text{Mg})/2}}$$
 , where Na and Mg are in milliequivalent per liter (me/l)

Ca_x is a modified Ca value calculated using Table 3-2 contained in *Irrigation with Reclaimed Municipal Wastewater, A Guidance Manual*.

Note: MGD = million gallons per day
mg/L = milligrams per liter
MPN/ 100 mL = Most Probable Number per 100 milliliters

2. Prior to commencement of the discharge and during periods when no discharge is occurring, the discharger shall submit a brief report in the monitoring report describing the status of the operation.
3. Flowrate shall be representative of the tertiary flow discharged from each facility.
4. Chlorine concentrations shall be recorded by a continuous recording meter. Calculated CT (chlorine concentration (x) modal time contact time) values shall be calculated at least monthly. The modal contact time must be at least 90 minutes based on peak daily design flow.
5. Samples for coliform bacteria shall be collected at least daily and at a time when waste water characteristics are most demanding on the treatment facilities and disinfection procedures.
6. Turbidity analysis shall be performed by a continuous recording turbidimeter, monthly reporting must include a calculation of the 95th percentile effluent turbidity for the month.
7. The discharger is not required to duplicate any metals analysis required under another Order but may submit a copy of the appropriate metals analysis results in compliance with this Order.

C. POTABLE WATER MONITORING

SOCRA shall obtain a quarterly analysis that is representative of the potable waters supplied to its service area and report the results annually.

Table C

POTABLE WATER CONSTITUENTS

CONSTITUENT	UNIT
Total Dissolved Solids	mg/L
Chloride	mg/L
Sulfate	mg/L
Sodium	mg/L

The annual report shall provide the average incremental increase of these constituents of the effluent over the supply water.

D. GROUNDWATER MONITORING

1. SOCRA shall obtain representative samples of the groundwater semi-annually and analyze for the following constituents with results reported annually. The samples shall be taken from Santa Margarita Water District's Ortega Well No. 3 and Chiquita WRP Well and Trabuco Canyon Water District's Upper Well and Lower Well.

Table D

GROUNDWATER MONITORING

CONSTITUENT	UNITS
Total Dissolved Solids	mg/L
Chloride	mg/L
Sulfate	mg/L
Sodium	mg/L
Fluoride	mg/L
Boron	mg/L
Nitrate	mg/L
Total Phosphate	mg/L
Iron	mg/L
Manganese	mg/L

Any changes of monitoring well locations shall reported in the annual report and be subject to approval by the Executive Officer or designated representative.

2. Groundwater levels shall be measured at least semi-annually to monitor the basin storage. Level reporting shall be made during the constituent sampling episodes in section D.1, above.
3. The method of sample collection and a demonstration that the sample is representative of groundwater conditions shall be included with each report.

E. OSO CREEK BARRIER and UPPER OSO RESERVOIR MONITORING

If samples are collected from Oso Creek and Upper Oso Reservoir, SOCRA shall shall prepare and submit a summary of all TDS data and submit the summary annually.

F. RECYCLED WATER USE MONITORING

Member agencies shall submit a recycled water users summary report quarterly to SOCRA, containing the following information:

1. Recycled water use site summary information

The following information shall be submitted for each recycled water use site:

- Name of the recycled water use site
- Address of the recycled water use site
- Owner of the recycled water use facility
- Name of the recycled water use supervisor
- Phone number of the recycled water use supervisor
- Mailing address of the recycled water use supervisor, if different from site address
- Basin Plan name and number of hydrologic subarea underlying the recycled water use site
- Volume of recycled water delivered to the recycled water use site on a monthly basis
- Title 22 level of treated waste water approved by the Department of Health Services delivered to the recycled water use site.

2. Recycled water use summary information

- Total volume of recycled water supplied to all recycled water users for each month of the reporting period.
- Total number of recycled water use sites.

3. Recycled water use site inspections

Number of recycled water use site inspections conducted by discharger/producer staff and identification of sites inspected for the reporting period.

4. Recycled water user violations of the Rules and Regulations for Recycled Water Use.

SOCRA shall identify all recycled water users known by SOCRA to be in violation of the Rules and Regulations for Recycled Water Use. The

report shall include a description of the noncompliance and its cause, including the period of noncompliance; and if the noncompliance has not been corrected, the anticipated time it is expected to continue and steps taken or planned to reduce, eliminate, and prevent recurrence of the noncompliance. This report shall be submitted to the Regional Board Executive Officer monthly.

SOCRA shall submit an annual Recycled Water Use Summary report that includes all reports submitted by member agencies, in an electronic format, and a written summary of the total volume of recycled water produced by each member, totalled monthly and annually and total recycled water use in each hydrologic subarea.

G. SEWAGE SOLIDS

SOCRA shall certify that each member agency has satisfied the requirements of Section D, Biosolids Specifications, of Order No. 97-52. This certification shall be submitted with the annual report.

H. CT REPORTING

A log of the monthly CT determinations shall be maintained by each producer of tertiary effluent and shall be submitted to SOCRA annually. SOCRA shall submit these CT logs with the annual report.

I. REPORTING

1. Member agencies shall report the following to the Regional Board Executive Officer and SOCRA within 24 hours:

Failure of chlorination equipment,
Effluent coliform MPN greater than 240/100 mL
Turbidity greater than 10 NTU
CT less than 300 (mg/L) (min)

2. Member agencies shall submit monitoring reports and recycled water use monitoring reports to SOCRA in accordance with the following schedule:

MEMBER AGENCY REPORTING FREQUENCY

REPORTING FREQUENCY	REPORT PERIOD	REPORT DUE
Monthly	January, February, March, April, May, June, July, August, September, October, November, December	By the 15th day of the fol- lowing month
Quarterly	January - March April - June July - September October - December	April 15 July 15 October 15 January 15
Annually	January - December	January 15

All reports shall be submitted to:

South Orange County Reclamation Authority
30290 Rancho Viejo Road
San Juan Capistrano, CA 92675

3. SOCRA shall submit reports to the Regional Board Executive Officer in accordance with the following schedule:

SOCRA REPORTING FREQUENCY

REPORTING FREQUENCY	REPORT DUE
Monthly	By the 30th day of the fol- lowing month
Annually	January 30

Monthly reports shall consist of a written summary of all exceedances of Discharge Specifications, Section A, of this Order, any violations reported pursuant to Recycled Water Use Monitoring, Section E.4, above and of any incidence reported pursuant to I.1, above.

October 15, 1997

Annual reports shall include accumulated monthly, quarterly and annual monitoring reports, and accumulated Recycled Water Use Reports, for each member agency. These accumulated reports may be submitted in an electronic format. The annual report shall also have a written summary of all violations not reported previously, the results of Potable Water Monitoring, Section C; Groundwater Monitoring Section D; Oso Creek Barrier and Upper Oso Reservoir Monitoring, Section E; Sewage Solids, Section G and CT Reporting, Section H.

Reports submitted in electronic format should be on a 3.5" computer diskette, formatted for IBM, and compatible with Excel, Wordperfect or Word for Windows format, as appropriate.

All reports shall be submitted by SOCRA to:

Executive Officer
California Regional Water Quality Control Board
San Diego Region
9771 Clairemont Mesa Blvd., Suite A
San Diego, California 92124-1324
Attn: Groundwater Unit

Ordered By:


John H. Robertus
Executive Officer
October 15, 1997